



Sesi 2: UI & Navigation dengan Flutter



Overview Sesi

Durasi: 90 menit (1.5 jam) | **Branch:** 01-hello-nylo → 02-home-page → 03-add-page → 04-detail-page | **Tujuan:** Belajar Flutter widgets dasar dan navigasi antar halaman



Agenda Sesi

Bagian 1: Flutter Widgets Fundamentals (30 menit)

- ✓ Pengenalan widget tree di Flutter
- ✓ Widget-widget dasar: Container, Column, Row, Text, Icon
- ✓ ListView untuk menampilkan list data
- ✓ Card untuk container dengan shadow
- ✓ Button widgets: ElevatedButton, IconButton, FloatingActionButton

Bagian 2: Build Home Page (30 menit)

- ✓ Generate HomePage dengan Metro CLI
- ✓ Build UI list todo dengan data static
- ✓ Styling cards dan layout

Bagian 3: Multi-Page Navigation (30 menit)

- ✓ Generate AddTaskPage dan DetailTaskPage
 - ✓ Setup routes di Nylo
 - ✓ Navigasi antar halaman dengan routeTo()
 - ✓ Passing data antar halaman
-

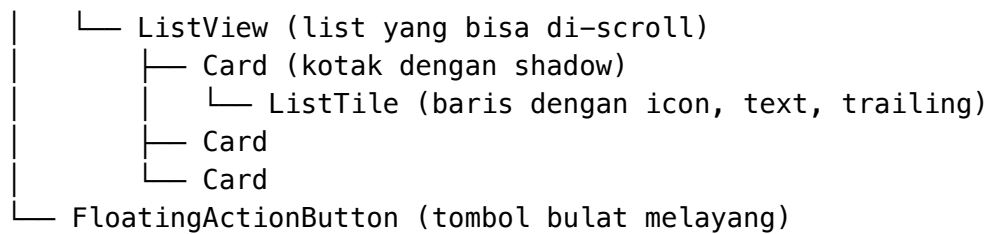


Bagian 1: Flutter Widgets Fundamentals (30 menit)

1.1 Widget Tree Concept

Flutter itu seperti menyusun balok LEGO. Setiap balok adalah **widget**, dan kita susun widget-widget ini untuk membuat tampilan aplikasi.

```
Scaffold (kerangka utama)
├─ AppBar (header di atas)
└─ Body
```



Analogi sederhana: - **Scaffold** = Rumah (kerangka utama) - **AppBar** = Atap rumah (bagian atas) - **Body** = Ruangan dalam rumah - **ListView** = Rak yang bisa di-scroll - **Card** = Kotak penyimpanan di rak

1.2 Widget-Widget Dasar

Container

Widget paling fleksibel untuk layout. Seperti kotak yang bisa diatur ukuran, warna, padding, margin.

```
Container(
  width: 200,           // lebar
  height: 100,          // tinggi
  padding: EdgeInsets.all(16), // jarak dalam
  margin: EdgeInsets.all(8),  // jarak luar
  decoration: BoxDecoration(
    color: Colors.blue,      // warna background
    borderRadius: BorderRadius.circular(12), // sudut melengkung
  ),
  child: Text('Hello'), // isi container
)
```

Kapan pakai Container? - Kalau butuh background warna - Kalau butuh padding/margin - Kalau butuh border radius (sudut melengkung)

Column & Row

Column = Susun widget secara **vertikal** (atas ke bawah) **Row** = Susun widget secara **horizontal** (kiri ke kanan)

```
// Column - susun vertical
Column(
  children: [
    Text('Item 1'),
    Text('Item 2'),
    Text('Item 3'),
  ],
)
```

```
// Row - susun horizontal
Row(
  children: [
    Icon(Icons.star),
    Text('Rating'),
    Text('4.5'),
  ],
)
```

```
],  
)
```

Alignment:

```
Column(  
  mainAxisAlignment: MainAxisAlignment.center, // posisi vertikal  
  crossAxisAlignment: CrossAxisAlignment.start, // posisi horizontal  
  children: [...],  
)
```

Analogi: - mainAxisAlignment = Posisi sepanjang arah utama (vertikal untuk Column, horizontal untuk Row) - crossAxisAlignment = Posisi tegak lurus arah utama

ListView

Untuk menampilkan list data yang bisa di-scroll. Ada 2 cara:

1. ListView dengan children (untuk data sedikit/fixed):

```
ListView(  
  children: [  
    Text('Item 1'),  
    Text('Item 2'),  
    Text('Item 3'),  
  ],  
)
```

2. ListView.builder (untuk data banyak/dynamic):

```
ListView.builder(  
  itemCount: 10, // jumlah item  
  itemBuilder: (context, index) {  
    // Widget yang akan dibuat untuk setiap item  
    return Text('Item $index');  
  },  
)
```

Kapan pakai ListView.builder? - Data banyak (10+ item) - Data dari variable/list - Performance lebih baik karena hanya build widget yang terlihat di layar

Card

Widget untuk membuat container dengan shadow (efek mengambang).

```
Card(  
  elevation: 4, // tinggi shadow (0-20)  
  margin: EdgeInsets.all(8),  
  child: ListTile(  
    leading: Icon(Icons.check_circle),  
    title: Text('Task Title'),  
    subtitle: Text('Task description here'),  
    trailing: Icon(Icons.more_vert),  
  ),  
)
```

```
),  
)
```

Kombinasi Card + ListTile sangat umum untuk membuat list item yang rapi!

Button Widgets

1. ElevatedButton - Button dengan shadow dan background warna:

```
ElevatedButton(  
  onPressed: () {  
    print('Button diklik!');  
  },  
  child: Text('Save'),  
)
```

2. IconButton - Button berbentuk icon saja:

```
IconButton(  
  icon: Icon(Icons.delete),  
  onPressed: () {  
    print('Delete clicked');  
  },  
)
```

3. FloatingActionButton - Button bulat yang melayang (biasa di kanan bawah):

```
FloatingActionButton(  
  onPressed: () {  
    print('FAB clicked');  
  },  
  child: Icon(Icons.add),  
)
```

TextField

Widget untuk input text dari user.

```
TextField(  
  decoration: InputDecoration(  
    labelText: 'Task Title',  
    hintText: 'Enter task title here',  
    border: OutlineInputBorder(),  
    prefixIcon: Icon(Icons.edit),  
  ),  
  onChanged: (value) {  
    print('User ketik: $value');  
  },  
)
```

Properties penting: - `labelText` = Label yang muncul di atas - `hintText` = Placeholder text abu-abu - `border` = Border style - `onChanged` = Function yang dipanggil setiap kali user ketik

1.3 Hands-on: Coba Widget Dasar

Mari coba modifikasi `home_page.dart` bawaan Nylo untuk eksplorasi widgets.

Langkah 1: Buka file `home_page.dart`

Path: lib/resources/pages/home_page.dart

Langkah 2: Ganti body dengan widget experiments:

```
// lib/resources/pages/home_page.dart
import 'package:flutter/material.dart';
import 'package:nylo_framework/nylo_framework.dart';
import '/app/controllers/home_controller.dart';

class HomePage extends NyStatefulWidget<HomeController> {
  static RouteView path = ("/home", (_) => HomePage());

  HomePage({super.key}) : super(child: () => _HomePageState());
}

class _HomePageState extends NyState<HomePage> {
  @override
  init() async {
    super.init();
  }

  @override
  Widget view(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text('Widget Playground'),
      ),
      body: ListView(
        padding: EdgeInsets.all(16),
        children: [
          // Container example
          _buildContainerExample(),
          SizedBox(height: 16),

          // Row example
          _buildRowExample(),
          SizedBox(height: 16),

          // Card example
          _buildCardExample(),
          SizedBox(height: 16),

          // TextField example
          _buildTextFieldExample(),
        ],
      ),
    );
  }
}
```

```

Widget _buildContainerExample() {
  return Container(
    padding: EdgeInsets.all(16),
    decoration: BoxDecoration(
      color: Colors.blue.shade100,
      borderRadius: BorderRadius.circular(12),
    ),
    child: Text(
      'Ini Container dengan background biru',
      style: TextStyle(fontSize: 16),
    ),
  );
}

Widget _buildRowExample() {
  return Row(
    mainAxisAlignment: MainAxisAlignment.spaceAround,
    children: [
      Icon(Icons.star, color: Colors.orange, size: 32),
      Icon(Icons.favorite, color: Colors.red, size: 32),
      Icon(Icons.thumb_up, color: Colors.blue, size: 32),
    ],
  );
}

Widget _buildCardExample() {
  return Card(
    elevation: 4,
    child: ListTile(
      leading: CircleAvatar(
        child: Icon(Icons.check),
      ),
      title: Text('Contoh Card + ListTile'),
      subtitle: Text('Ini subtitle'),
      trailing: Icon(Icons.arrow_forward),
    ),
  );
}

Widget _buildTextFieldExample() {
  return TextField(
    decoration: InputDecoration(
      labelText: 'Coba ketik di sini',
      hintText: 'Placeholder text',
      border: OutlineInputBorder(),
      prefixIcon: Icon(Icons.edit),
    ),
  );
}

```

Langkah 3: Run aplikasi dan lihat hasilnya!

flutter run

💡 **Coba-coba sendiri:** - Ganti warna background Container - Tambah icon baru di Row
- Ubah text di Card - Tambahkan suffixIcon di TextField

📸 **Screenshot:** Ambil screenshot hasil percobaan Anda dan bandingkan dengan teman!

🏠 **Bagian 2: Build Home Page ToDo List (30 menit)**

Sekarang kita akan membuat halaman utama aplikasi ToDo List dengan data static (hardcode dulu).

2.1 Generate HomePage Baru

```
# Generate page baru untuk todo list
dart run nylo_framework:main make:page todo_home_page
```

Output:

- ✓ Created lib/resources/pages/todo_home_page.dart
 - ✓ Created lib/app/controllers/todo_home_controller.dart
-

2.2 Setup Route untuk TodoHomePage

Edit file: `lib/routes/router.dart`

```
import 'package:nylo_framework/nylo_framework.dart';
import '/resources/pages/home_page.dart';
import '/resources/pages/todo_home_page.dart';

appRouter() => nyRoutes((router) {
  router.add(HomePage.path);

  // Route baru untuk todo home
  router.add(TodoHomePage.path);
});
```

2.3 Update Main App untuk Start di TodoHomePage

Edit file: `lib/app/providers/route_provider.dart`

Cari bagian `initialRoute` dan ganti ke `/todo-home`:

```
// lib/app/providers/route_provider.dart
import 'package:flutter/material.dart';
import 'package:nylo_framework/nylo_framework.dart';

class RouteProvider implements NyProvider {
  @override
  boot(Nylo nylo) async {
    return nylo;
  }
}
```

```

}

@override
afterBoot(Nylo nylo) async {
  NyNavigator.instance.router = nylo.router;

  // Ubah initial route ke todo home page
  NyNavigator.instance.initialRoute = '/todo-home'; // <-- UBAH INI
}
}

```

2.4 Build TodoHomePage UI

Edit file: `lib/resources/pages/todo_home_page.dart`

```

import 'package:flutter/material.dart';
import 'package:nylo_framework/nylo_framework.dart';
import '/app/controllers/todo_home_controller.dart';

class TodoHomePage extends NyStatefulWidget<TodoHomeController> {
  static RouteView path = ("/todo-home", (_) => TodoHomePage());

  TodoHomePage({super.key}) : super(child: () =>
    _TodoHomePageState());
}

class _TodoHomePageState extends NyState<TodoHomePage> {
  @override
  init() async {
    super.init();
  }

  @override
  Widget view(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text('My Todo List'),
        backgroundColor: Colors.blue,
        foregroundColor: Colors.white,
      ),
      body: _buildTaskList(),
      floatingActionButton: FloatingActionButton(
        onPressed: () {
          // TODO: Nanti akan navigate ke Add Task Page
          print('FAB diklik!');
        },
        child: Icon(Icons.add),
      ),
    );
  }

  Widget _buildTaskList() {
    // Data static untuk sementara (hardcode)
    List<Map<String, dynamic>> tasks = [

```



```

{
  'id': '1',
  'title': 'Belajar Flutter',
  'description': 'Selesaikan tutorial Flutter basics',
  'isCompleted': false,
},
{
  'id': '2',
  'title': 'Setup Project Nylo',
  'description': 'Install dan setup Nylo framework',
  'isCompleted': true,
},
{
  'id': '3',
  'title': 'Build ToDo App',
  'description': 'Buat aplikasi ToDo sederhana',
  'isCompleted': false,
},
];

return ListView.builder(
  padding: EdgeInsets.all(16),
  itemCount: tasks.length,
  itemBuilder: (context, index) {
    final task = tasks[index];
    return _buildTaskCard(task);
  },
);
}

Widget _buildTaskCard(Map<String, dynamic> task) {
  bool isCompleted = task['isCompleted'] ?? false;

  return Card(
    elevation: 2,
    margin: EdgeInsets.only(bottom: 12),
    child: ListTile(
      // Icon di kiri
      leading: CircleAvatar(
        backgroundColor: isCompleted ? Colors.green : Colors.orange,
        child: Icon(
          isCompleted ? Icons.check : Icons.circle_outlined,
          color: Colors.white,
        ),
      ),
      // Title & description
      title: Text(
        task['title'],
        style: TextStyle(
          fontSize: 16,
          fontWeight: FontWeight.bold,
          // Coret text jika sudah selesai
          decoration: isCompleted ? TextDecoration.lineThrough :
            null,

```

```

    ),
  ),
  subtitle: Text(
    task['description'],
    style: TextStyle(fontSize: 14, color: Colors.grey),
  ),

  // Icon di kanan
  trailing: Icon(Icons.arrow_forward_ios, size: 16),

  // Action ketika card diklik
  onTap: () {
    print('Task "${task['title']}" diklik!');
    // TODO: Nanti akan navigate ke Detail Page
  },
),
);
}
}

```

🔥 **Run aplikasi dan lihat hasilnya!**

```

flutter run
# atau tekan R di terminal untuk hot reload

```

📋 **Checklist:** - [] Muncul AppBar dengan title “My Todo List” - [] Ada 3 task cards di list - [] Task ke-2 sudah selesai (icon hijau, text coret) - [] FAB (tombol +) muncul di kanan bawah - [] Ketika card diklik, muncul print di console

2.5 Improve UI dengan Styling

Mari tambahkan sedikit styling agar lebih menarik!

Tambahkan section stats di atas list:

@override

```

Widget view(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: Text('My Todo List'),
      backgroundColor: Colors.blue,
      foregroundColor: Colors.white,
    ),
    body: Column(
      children: [
        _buildStatsCard(), // <-- Tambah stats card
        Expanded(child: _buildTaskList()),
      ],
    ),
    floatingActionButton: FloatingActionButton(
      onPressed: () {
        print('FAB diklik!');
      },
      child: Icon(Icons.add),
    ),
  );
}

```

```

    ),
  );
}

Widget _buildStatsCard() {
  // Hitung dari data static
  int totalTasks = 3;
  int completedTasks = 1;
  int pendingTasks = 2;

  return Container(
    margin: EdgeInsets.all(16),
    padding: EdgeInsets.all(16),
    decoration: BoxDecoration(
      color: Colors.blue.shade50,
      borderRadius: BorderRadius.circular(12),
    ),
    child: Row(
      mainAxisAlignment: MainAxisAlignment.spaceAround,
      children: [
        _buildStatItem('Total', totalTasks, Colors.blue),
        _buildStatItem('Done', completedTasks, Colors.green),
        _buildStatItem('Pending', pendingTasks, Colors.orange),
      ],
    ),
  );
}

Widget _buildStatItem(String label, int count, Color color) {
  return Column(
    children: [
      Text(
        count.toString(),
        style: TextStyle(
          fontSize: 28,
          fontWeight: FontWeight.bold,
          color: color,
        ),
      ),
      SizedBox(height: 4),
      Text(
        label,
        style: TextStyle(
          fontSize: 14,
          color: Colors.grey.shade700,
        ),
      ),
    ],
  );
}

```

Hot reload dan lihat stats card muncul di atas list!



Bagian 3: Multi-Page Navigation (30 menit)

Sekarang kita akan buat 2 halaman baru: 1. **AddTaskPage** - Form untuk tambah task baru 2. **DetailTaskPage** - Detail task yang diklik

3.1 Generate AddTaskPage

```
dart run nylo_framework:main make:page add_task_page
```

Output:

- ✓ Created lib/resources/pages/add_task_page.dart
 - ✓ Created lib/app/controllers/add_task_controller.dart
-

3.2 Build AddTaskPage UI

Edit file: **lib/resources/pages/add_task_page.dart**

```
import 'package:flutter/material.dart';
import 'package:nylo_framework/nylo_framework.dart';
import '/app/controllers/add_task_controller.dart';

class AddTaskPage extends NyStatefulWidget<AddTaskController> {
  static RouteView path = ("/add-task", (_) => AddTaskPage());

  AddTaskPage({super.key}) : super(child: () => _AddTaskPageState());
}

class _AddTaskPageState extends NyState<AddTaskPage> {
  // Controller untuk TextField
  final TextEditingController titleController =
    TextEditingController();
  final TextEditingController descriptionController =
    TextEditingController();

  @override
  void dispose() {
    // Bersihkan controller saat page ditutup
    titleController.dispose();
    descriptionController.dispose();
    super.dispose();
  }

  @override
  Widget view(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text('Add New Task'),
        backgroundColor: Colors.blue,
        foregroundColor: Colors.white,
      ),
      body: SingleChildScrollView(
        padding: EdgeInsets.all(16),
```

```

child: Column(
  crossAxisAlignment: CrossAxisAlignment.start,
  children: [
    // Title input
    TextField(
      controller: titleController,
      decoration: InputDecoration(
        labelText: 'Task Title',
        hintText: 'e.g. Belajar Flutter',
        border: OutlineInputBorder(),
        prefixIcon: Icon(Icons.title),
      ),
    ),
    SizedBox(height: 16),

    // Description input
    TextField(
      controller: descriptionController,
      maxLines: 4,
      decoration: InputDecoration(
        labelText: 'Description (optional)',
        hintText: 'Describe your task here...',
        border: OutlineInputBorder(),
        prefixIcon: Icon(Icons.description),
      ),
    ),
    SizedBox(height: 24),

    // Save button
    SizedBox(
      width: double.infinity,
      height: 50,
      child: ElevatedButton(
        onPressed: _handleSave,
        style: ElevatedButton.styleFrom(
          backgroundColor: Colors.blue,
          foregroundColor: Colors.white,
        ),
        child: Text(
          'Save Task',
          style: TextStyle(fontSize: 16),
        ),
      ),
    ),
  ],
),
);
}

void _handleSave() {
  String title = titleController.text.trim();
  String description = descriptionController.text.trim();

  // Validasi

```

```

    if (title.isEmpty) {
      // Show error message
      showToastNotification(
        context,
        title: 'Error',
        description: 'Task title tidak boleh kosong!',
        style: ToastNotificationStyleType.DANGER,
      );
      return;
    }

    // Print untuk testing (nanti akan save ke storage)
    print('Saving task:');
    print('Title: $title');
    print('Description: $description');

    // Show success message
    showToastNotification(
      context,
      title: 'Success',
      description: 'Task berhasil ditambahkan!',
      style: ToastNotificationStyleType.SUCCESS,
    );

    // Kembali ke halaman sebelumnya setelah 1 detik
    Future.delayed(Duration(seconds: 1), () {
      pop(context);
    });
  }
}

```

3.3 Generate DetailTaskPage

```
dart run nylo_framework:main make:page detail_task_page
```

3.4 Build DetailTaskPage UI

Edit file: `lib/resources/pages/detail_task_page.dart`

```

import 'package:flutter/material.dart';
import 'package:nylo_framework/nylo_framework.dart';
import '/app/controllers/detail_task_controller.dart';

class DetailTaskPage extends NyStatefulWidget<DetailTaskController> {
  static RouteView path = ("/detail-task", (_) => DetailTaskPage());

  DetailTaskPage({super.key}) : super(child: () =>
    _DetailTaskPageState());
}

class _DetailTaskPageState extends NyState<DetailTaskPage> {
  // Data task yang diterima dari halaman sebelumnya

```

```
Map<String, dynamic>? task;
```

```
@override
```

```
init() async {  
  super.init();  
  // Ambil data task dari route arguments  
  task = widget.data();  
}
```

```
@override
```

```
Widget view(BuildContext context) {  
  // Jika task null, tampilkan error  
  if (task == null) {  
    return Scaffold(  
      appBar: AppBar(title: Text('Error')),  
      body: Center(child: Text('Task not found')),  
    );  
  }  
}
```

```
bool isCompleted = task!['isCompleted'] ?? false;
```

```
return Scaffold(  
  appBar: AppBar(  
    title: Text('Task Detail'),  
    backgroundColor: Colors.blue,  
    foregroundColor: Colors.white,  
    actions: [  
      IconButton(  
        icon: Icon(Icons.edit),  
        onPressed: () {  
          print('Edit button diklik');  
          // TODO: Navigate ke edit page  
        },  
      ),  
      IconButton(  
        icon: Icon(Icons.delete),  
        onPressed: () {  
          print('Delete button diklik');  
          // TODO: Implement delete  
        },  
      ),  
    ],  
  ),  
  body: SingleChildScrollView(  
    padding: EdgeInsets.all(16),  
    child: Column(  
      crossAxisAlignment: CrossAxisAlignment.start,  
      children: [  
        // Status badge  
        Container(  
          padding: EdgeInsets.symmetric(horizontal: 12, vertical:  
6),  
          decoration: BoxDecoration(  
            color: isCompleted ? Colors.green : Colors.orange,  
            borderRadius: BorderRadius.circular(20),
```

```

    ),
    child: Row(
      mainAxisAlignment: MainAxisAlignment.min,
      children: [
        Icon(
          isCompleted ? Icons.check_circle : Icons.pending,
          color: Colors.white,
          size: 16,
        ),
        SizedBox(width: 4),
        Text(
          isCompleted ? 'Completed' : 'Pending',
          style: TextStyle(color: Colors.white),
        ),
      ],
    ),
  ),
  SizedBox(height: 20),

  // Title
  Text(
    'Title',
    style: TextStyle(
      fontSize: 12,
      color: Colors.grey,
      fontWeight: FontWeight.bold,
    ),
  ),
  SizedBox(height: 4),
  Text(
    task!['title'],
    style: TextStyle(
      fontSize: 24,
      fontWeight: FontWeight.bold,
    ),
  ),
  SizedBox(height: 20),

  // Description
  Text(
    'Description',
    style: TextStyle(
      fontSize: 12,
      color: Colors.grey,
      fontWeight: FontWeight.bold,
    ),
  ),
  SizedBox(height: 4),
  Text(
    task!['description'] ?? 'No description',
    style: TextStyle(fontSize: 16),
  ),
  SizedBox(height: 32),

  // Toggle complete button

```



```

        SizedBox(
          width: double.infinity,
          height: 50,
          child: ElevatedButton.icon(
            onPressed: () {
              print('Toggle complete: ${!isCompleted}');
              // TODO: Update task status
            },
            icon: Icon(isCompleted ? Icons.undo : Icons.check),
            label: Text(
              isCompleted ? 'Mark as Pending' : 'Mark as
Completed',
              style: TextStyle(fontSize: 16),
            ),
            style: ElevatedButton.styleFrom(
              backgroundColor: isCompleted ? Colors.orange :
Colors.green,
              foregroundColor: Colors.white,
            ),
          ),
        ),
      ],
    ),
  );
}
}

```

3.5 Setup Routes

Edit file: `lib/routes/router.dart`

```

import 'package:nylo_framework/nylo_framework.dart';
import '/resources/pages/home_page.dart';
import '/resources/pages/todo_home_page.dart';
import '/resources/pages/add_task_page.dart';
import '/resources/pages/detail_task_page.dart';

appRouter() => nyRoutes((router) {
  router.add(HomePage.path);
  router.add(TodoHomePage.path);
  router.add(AddTaskPage.path);
  router.add(DetailTaskPage.path);
});

```

3.6 Implement Navigation dari TodoHomePage

Sekarang kita hubungkan semua halaman dengan navigasi!

Edit file: `lib/resources/pages/todo_home_page.dart`

Update FAB action:

```
floatingActionButton: FloatingActionButton(
  onPressed: () {
    // Navigate ke Add Task Page
    routeTo('/add-task');
  },
  child: Icon(Icons.add),
),
```

Update onTap di task card:

```
Widget _buildTaskCard(Map<String, dynamic> task) {
  bool isCompleted = task['isCompleted'] ?? false;

  return Card(
    elevation: 2,
    margin: EdgeInsets.only(bottom: 12),
    child: ListTile(
      leading: CircleAvatar(
        backgroundColor: isCompleted ? Colors.green : Colors.orange,
        child: Icon(
          isCompleted ? Icons.check : Icons.circle_outlined,
          color: Colors.white,
        ),
      ),
      title: Text(
        task['title'],
        style: TextStyle(
          fontSize: 16,
          fontWeight: FontWeight.bold,
          decoration: isCompleted ? TextDecoration.lineThrough : null,
        ),
      ),
      subtitle: Text(
        task['description'],
        style: TextStyle(fontSize: 14, color: Colors.grey),
      ),
      trailing: Icon(Icons.arrow_forward_ios, size: 16),
      onTap: () {
        // Navigate ke Detail Page dengan passing data
        routeTo(
          '/detail-task',
          data: task, // <-- Pass task data ke halaman detail
        );
      },
    ),
  );
}
```

3.7 Testing Navigation

Test flow:

1. Run aplikasi

flutter run

2. Test FAB Navigation:

- Klik tombol + (FAB)
- Harus masuk ke Add Task Page
- Isi form dan klik Save
- Harus kembali ke Home Page dengan toast success

3. Test Card Navigation:

- Klik salah satu task card di home
- Harus masuk ke Detail Task Page
- Lihat detail task muncul
- Klik back button untuk kembali

 **Checklist:** - ☐ FAB navigate ke Add Task Page - ☐ Form validation bekerja (title wajib diisi) - ☐ Toast notification muncul saat save - ☐ Otomatis kembali ke home setelah save - ☐ Card navigate ke Detail Page dengan data yang benar - ☐ Detail page menampilkan title, description, dan status - ☐ Back button berfungsi di semua halaman

Bonus: Custom App Theme

Mari buat aplikasi lebih konsisten dengan custom theme!

Edit file: `lib/config/theme.dart`

```
import 'package:flutter/material.dart';
```

```
ThemeData getAppTheme() {  
  return ThemeData(  
    // Primary color  
    primarySwatch: Colors.blue,  
    primaryColor: Colors.blue,  
  
    // AppBar theme  
    appBarTheme: AppBarTheme(  
      backgroundColor: Colors.blue,  
      foregroundColor: Colors.white,  
      elevation: 2,  
      centerTitle: true,  
    ),  
  
    // FloatingActionButton theme  
    floatingActionButtonTheme: FloatingActionButtonThemeData(  
      backgroundColor: Colors.blue,  
      foregroundColor: Colors.white,  
    ),  
  
    // Card theme  
    cardTheme: CardTheme(  
      elevation: 2,  
      shape: RoundedRectangleBorder(  
        borderRadius: BorderRadius.circular(12),
```

```

    ),
  ),

  // Input decoration theme
  inputDecorationTheme: InputDecorationTheme(
    border: OutlineInputBorder(
      borderRadius: BorderRadius.circular(12),
    ),
    filled: true,
    fillColor: Colors.grey.shade50,
  ),

  // Button theme
  elevatedButtonTheme: ElevatedButtonThemeData(
    style: ElevatedButton.styleFrom(
      backgroundColor: Colors.blue,
      foregroundColor: Colors.white,
      shape: RoundedRectangleBorder(
        borderRadius: BorderRadius.circular(12),
      ),
    ),
  ),
);
}

```

Terapkan theme di MaterialApp:

Edit file: lib/app/providers/app_provider.dart

Cari bagian boot method dan pastikan theme sudah di-load:

```

@override
boot(Nylo nylo) async {
  nylo.addThemes([
    ThemeData.light(), // light theme
    ThemeData.dark(),  // dark theme
    getAppTheme(),     // custom app theme
  ]);

  nylo.themeId = 2; // Set ke index 2 (custom theme)

  return nylo;
}

```

Hot reload dan lihat perubahan styling di seluruh aplikasi!

Latihan Mandiri

Latihan 1: Empty State

Tambahkan empty state di TodoHomePage ketika list tasks kosong.

Hint:

```

if (tasks.isEmpty) {
  return Center(
    child: Column(
      mainAxisAlignment: MainAxisAlignment.center,
      children: [
        Icon(Icons.inbox, size: 64, color: Colors.grey),
        SizedBox(height: 16),
        Text('No tasks yet!'),
        Text('Tap + button to add your first task'),
      ],
    ),
  );
}

```

Latihan 2: Search Bar

Tambahkan search bar di AppBar untuk filter tasks.

Hint: - Gunakan TextField di AppBar - Filter tasks dengan `tasks.where((t) => t['title'].contains(query))`

Latihan 3: Confirmation Dialog

Tambahkan dialog konfirmasi sebelum delete di Detail Page.

Hint:

```

showDialog(
  context: context,
  builder: (context) => AlertDialog(
    title: Text('Delete Task?'),
    content: Text('Are you sure?'),
    actions: [
      TextButton(onPressed: () => pop(context), child:
        Text('Cancel')),
      TextButton(onPressed: () {
        // Delete logic
        pop(context);
      }, child: Text('Delete')),
    ],
  ),
);

```



Common Issues & Solutions

Issue 1: “Navigator operation requested with a context that does not include a Navigator”

Symptom: Error saat panggil `routeTo()` **Solution:** - Pastikan `appRouter()` sudah didaftarkan di `RouteProvider` - Pastikan semua pages sudah di-add di `router.dart` - Gunakan context dari `build()` method, bukan dari class level

Issue 2: “data() returns null”

Symptom: Data tidak sampai ke halaman tujuan **Solution:** - Pastikan pass data dengan parameter data: di `routeTo()` - Panggil `widget.data()` di `init()` method, bukan di `build()`

Issue 3: “Hot reload not working”

Symptom: Perubahan kode tidak muncul di app **Solution:** - Coba tekan R di terminal untuk hot reload - Jika tidak berhasil, tekan Shift+R untuk hot restart - Last resort: Stop app dan flutter run ulang

Issue 4: “TextField not clearing after submit”

Symptom: TextField masih berisi text setelah save **Solution:**

```
titleController.clear();  
descriptionController.clear();
```



Key Takeaways

- ✅ **Flutter Widgets:** - Flutter itu widget tree - semua adalah widget - Scaffold kerangka dasar page (AppBar + Body + FAB) - `ListView.builder` untuk list dinamis - Card + `ListTile` kombinasi umum untuk list items - TextField untuk input dari user
 - ✅ **Navigation di Nylo:** - Generate page dengan metro `make:page` - Daftarkan route di `router.dart` - Navigate dengan `routeTo('/path')` - Pass data dengan `routeTo('/path', data: {...})` - Terima data dengan `widget.data()`
 - ✅ **Best Practices:** - Pisahkan widget besar jadi method kecil (`_buildXxx()`) - Gunakan `TextEditingController` untuk TextField - Jangan lupa `dispose()` controller - Validasi input sebelum process - Berikan feedback ke user (toast, dialog)
-



Resources

- **Flutter Widgets Catalog:** <https://docs.flutter.dev/ui/widgets>
 - **Material Design Guidelines:** <https://m3.material.io/>
 - **Nylo Routing:** <https://nylo.dev/docs/6.x/router>
 - **Flutter Layout Cheatsheet:** <https://medium.com/flutter-community/flutter-layout-cheat-sheet-5363348d037e>
-



Navigation

← Sesi 1 - Hello Flutter with Nylo | Sesi 3 - State Management & Local Data →
